

In this train, we'll **uncover the essential role of data engineering**. We'll explore how data engineers build and maintain the intricate systems that keep data flowing smoothly, allowing organisations to **harness the power of data** for decision-making, innovation, and solving complex problems.

Learning Objectives

By the end of this lesson, you should be able to:

- Define data engineering, understand its significance in the data lifecycle, and its role in enabling data-driven decision-making.
- Identify the key processes involved in designing and maintaining data infrastructure.
- Explain how data engineering supports effective data analysis and its **impact on various industries**.

What Is Data Engineering?

Data engineering is a branch of data science that focuses on **designing, building, and maintaining systems** for storing, processing, and analysing data. The goal is to ensure that data is **available, reliable, and accessible** for data analysts, data scientists, and other stakeholders who rely on this data for insights and decision-making.

Key concepts to understand:

Data Lifecycle

A data lifecycle illustrates how data, in all its various forms, conceptually flows through its **lifecycle of usefulness**. It refers to the **different stages a unit of data undergoes**, from initial collection to when it's no longer considered useful and deleted.

Data Pipelines

Think of a data pipeline as a **series of steps** that transform raw data into something useful. Data engineers design these pipelines to **move data from one place to another**, transforming and cleansing it along the way.

Data Infrastructure

This encompasses the **tools, technologies, and systems** used to store and manage data. Data infrastructure is the foundation that supports data engineering tasks.

Data Workflow

A data workflow is a broader concept that includes not only the **data pipeline** but also the **orchestration and automation** of tasks involved in data processing. Workflows often manage how and when different data pipelines are triggered, **coordinate dependencies** between different pipelines, and handle **error checking, monitoring, and reporting**.

Reflect:

Reflect on your day-to-day life. How often do you interact with data? What apps or tools do you use that rely on data? Think about what would happen if these tools didn't work smoothly. That's the challenge data engineers tackle.

In today's world, data is as valuable as oil once was. Companies use data to understand their customers, optimise their operations, and make informed decisions. Data engineering makes this possible by ensuring that data is available, reliable, and easy to use.

Key processes in data engineering

Let's delve deeper into the journey that data takes and the key processes data engineers use to manage it. Each stage represents an essential part of the flow, ensuring data remains clean, accessible, and ready for analysis.

Data Ingestion

Data ingestion is the first step in the journey, where **raw data is collected from multiple sources**, such as databases, APIs, and IoT devices.

Modes of ingestion:

1. **Batch**: In this mode, data is **collected and processed in groups** or "batches" at scheduled intervals. It's like waiting until the end of each day to gather all the new information and then process it at once. This approach is suitable when real-time processing isn't required, and it can **handle large volumes of data** efficiently.
2. **Streaming**: Streaming ingestion involves the **continuous, real-time flow of data**. Instead of waiting for intervals, data is processed as soon as it's generated. This mode is essential for applications that require **instant data updates** and quick responses, such as monitoring social media feeds or tracking IoT sensor data.

Prompt:

Think about your favourite streaming service. How do they ensure that new songs, shows, or movies are available instantly? Which mode do you think they use, and why?

Once data is collected, it must be **cleaned and standardised**. Transformation ensures that data is in a usable format for analysis.

Methods of transformation:

1. **ETL (Extract, Transform, Load)** : In the ETL process, data is first **extracted** from various source systems, then **transformed** into a desired format, structure, or schema, and finally **loaded** into a target data storage system, such as a data warehouse. This approach is ideal when data **quality and consistency are critical** and data needs to be processed and validated before being stored.
2. **ELT (Extract, Load, Transform)** : In ELT, data is **extracted** from source systems and **loaded** directly into a data lake or other storage system in its raw form. The **transformation** occurs later, often on an as-needed basis. This approach is well-suited for exploratory data analysis or machine learning, where **different teams might need to transform the data in unique ways**.

Prompt:

Consider a social media company that stores a vast amount of unstructured user data, including text, images, and videos. Analysts frequently experiment with the raw data to discover trends and insights. Would you choose **ETL or ELT** for this scenario? How might your choice impact data accessibility and processing time?

Transformed data is **stored in structured systems for easy access**. Depending on the data, it might be stored in databases, data warehouses, or data lakes.

Types of storage:

1. **Databases** : Databases are designed for storing **structured data** in an organised, efficient, and accessible way. They are often used to manage transactional data, such as customer records, inventory levels, or financial transactions. Data is stored in tables with **defined rows and columns**, making it easy to search, update, and retrieve information quickly.
2. **Data warehouses** : Data warehouses are **optimised for the analysis and reporting of large volumes of historical data**. They store aggregated, structured data from multiple sources in a centralised location, allowing organisations to generate insights through business intelligence tools.
3. **Data lakes** : Data lakes are designed to store vast amounts of **raw data in its native format**, whether structured, semi-structured, or unstructured. Unlike databases or data warehouses, data lakes **do not require a predefined schema**. This flexibility allows organisations to store everything from structured logs and databases to semi-structured JSON files and unstructured text, video, or image files.

Prompt:

Imagine you need to decide where to store three types of data for an e-commerce platform: detailed customer purchase records, summarised sales reports for business analysis, and raw user behaviour logs generated in real-time.

Based on factors like data structure, analysis needs, and real-time access, which storage method would you choose for each type and why?

Data orchestration is the process of **coordinating and managing data workflows** to ensure data moves smoothly and efficiently from one stage to another. In data engineering, this involves **automating tasks** like data ingestion, transformation, and loading, while also **monitoring these tasks** to ensure they occur in the correct sequence and on schedule.

Key functions of data orchestration:

1. **Scheduling**: Setting specific times for tasks to run, whether hourly, daily, or based on an event trigger (like the arrival of new data).
2. **Dependency management**: Ensuring tasks are executed in the correct order, with downstream tasks waiting for upstream tasks to complete.
3. **Error handling and recovery**: Detecting errors in the workflow and triggering corrective actions or alerts to prevent data flow disruptions.
4. **Monitoring and logging**: Continuously tracking the workflow's status and generating logs for auditing and troubleshooting.

Data quality assurance is a critical step in the data engineering process. It involves ensuring that the data moving through pipelines is **accurate, consistent, complete, and reliable**. Poor data quality can lead to incorrect analysis, faulty decisions, and ultimately, negative impacts on business operations. Thus, data engineers implement various checks and processes to **maintain high-quality data**.

Data quality is measured and assured across **six dimensions**:



The Role of Data Engineering in Data Analysis

The journey culminates in data analysis, where transformed, stored, and validated data can finally be used to gain insights, make decisions, and drive innovation.

Data engineering ensures that data analysts and scientists can access data that is:

Accessible

Data engineers ensure data is stored in easily accessible formats and organised systems by **optimising storage solutions and creating data catalogues**, so that data analysts and scientists can quickly retrieve and work with the data they need.

Reliable

Data engineers ensure data reliability by **implementing quality checks and monitoring mechanisms**, so that data analysts and scientists can trust that the data they are using is accurate, consistent, and free from errors.

Scalable

Data engineers ensure scalability by **designing systems that can grow with increasing data volumes** and user demand, so that data analysts and scientists can continue to perform complex analyses on large datasets as the organisation expands.

Final Thoughts

Reflect on how data engineering affects our everyday lives and the vast potential it holds for shaping the future. Consider where else data engineering might have an impact that you hadn't thought of before.